12. Plane Isometries

1. (See the answer in the text.)

2.

	P	R
P	P	R
R	R	P

3. (See the answer in the text.)

4. A figure with a one-element group of plane symmetries.



5. A figure with a two-element group of plane symmetries.

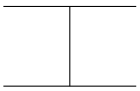


6. A figure with a three-element group of plane symmetries.



7. (See the answer in the text.)

8. A figure with a four-element group of plane symmetries, isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.



9. (See the answer in the text.)

10. Rotations and reflections can have fixed points. A translation slides all points by the same amount, and a glide reflection moves all points the same distance.

11. A rotations is the only type with just one fixed point.

12. No plane isometry has exactly two fixed points. If P and Q are left fixed, so are all points on the line through these two points.

13. Only the identity and reflections have an infinite number of fixed points.

14. If P, Q, and R are three non collinear points, then three circles with centers at P, Q, and R have at most one point in common. Namely, two circles intersect in two points, and if the center of the third circle does not lie on the line through the centers of the first two, then it can't pass through both points of intersection of the first two. An isometry ϕ that leaves P, Q, and R fixed must leave every other point S fixed because it must preserve its distance to P, Q, and R, so that both S and $\phi(S)$ must be the unique points of intersection of three circles with P, Q, and R as centers and the appropriate radii.

15. If $\phi(P_i) = \psi(P_i)$ for $i = 1, 2$, and 3, then $\phi^{-1}(\psi(P_i)) = P_i$ for $i = 1, 2$, and 3. Thus by Exercise 14, $\phi^{-1}\psi = \iota$, the identity map, so $\psi = \phi$.

16. No, the product of two rotations (about different points) may be a translation, so the set of rotations is not closed under multiplication.

17. (See the answer in the text.)

18. Yes, they do form a subgroup. Think of the fixed point as the origin in the plane of complex numbers. Rotations about that point correspond to multiplying by complex numbers z such that $|z| = 1$. The set $U = \{z \in \mathbb{C} \mid |z| = 1\}$ is a group under multiplication, and the multiplication corresponds to function composition of rotations. The number 1 corresponds to the identity map.

19. (See the answer in the text.)

20. No, the product of two glide reflections is orientation preserving, and hence is not a glide reflection.

21. (See the answer in the text.)

22. Because G is finite, it can contain no translations, so the orientation preserving isometries in G consist of the rotations in G and the identity map. Because the product of two orientation preserving isometries is orientation preserving, we see that the set H of all orientation preserving isometries in G is closed under multiplication (function composition). Because the inverse of a rotation is also a rotation, we see that H contains the inverse of each element, and is thus a subgroup of G. If $H \neq G$, let μ be an element of G that is not in H. If σ is another element of G not in H, then $\mu^{-1}\sigma \in H$, because the product of two orientation reversing isometries is order preserving. Thus $\sigma \in \mu H$. This shows that the coset μH contains all elements of G that are not in H. Because $|\mu H| = |H|$, we see that in this case $|G| = 2|H|$.

23. We can consider all the rotations in G to be clockwise. Let ρ be the rotation in G which rotates the plane clockwise through the smallest positive angle. Such a rotation exists because G is a finite group. We claim that G is cyclic, generated by ρ. Let α be the angle of rotation for ρ. Let σ be another rotation in G with angle of rotation β. Write $\beta = q\alpha + \theta$, according to the division algorithm. Then $\theta = \beta - q\alpha$, and the isometry $\rho^{-q}\sigma$ rotates the plane through the angle θ. By the division algorithm, either $\theta = 0$ or $0 < \theta < \alpha$. Because $0 < \theta < \alpha$ is impossible by our choice of α as the smallest nonzero angle of rotation, we see that $\theta = 0$. Hence $\beta = q\alpha$, so $\sigma = \rho^q$, showing that G is cyclic and generated by ρ.

24. **a.** No **b.** No **c.** No **d.** No **e.** $\mathbb{Z}$

25. **a.** No **b.** No **c.** Yes **d.** No **e.** D_∞

26. **a.** No **b.** Yes **c.** No **d.** No **e.** $\mathbb{Z} \times \mathbb{Z}_2$

27. **a.** Yes **b.** No **c.** No **d.** No **e.** D_∞

28. **a.** Yes **b.** Yes **c.** Yes **d.** No **e.** $D_\infty \times \mathbb{Z}_2$

29. **a.** No **b.** No **c.** No **d.** Yes **e.** $\mathbb{Z}$

30. **a.** Yes **b.** No **c.** Yes **d.** Yes **e.** D_∞

31. **a.** Yes, 90° and 180° **b.** Yes **c.** No

32. **a.** Yes, 180° **b.** Yes **c.** No